HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented Python data pipelines for ETL processes, transforming and loading data for reporting and analytical use cases, processing approximately 1.5 TB of data weekly.
- Developed and maintained Java Spring Boot REST APIs for internal systems, handling over 5,000 requests per minute during peak operational hours.
- Managed production support for enterprise applications, including P1/P2 incident response and SLA adherence, resolving an average of 15 critical issues per month.
- Built interactive dashboards and reports using **Power BI**, translating complex datasets into actionable insights for business stakeholders, delivering 10 key reports to management.
- Wrote and optimized complex SQL queries across PostgreSQL and Oracle databases for data validation and ad-hoc analysis, supporting the generation of 20+ unique analytical queries per sprint.
- Containerized multiple microservices using **Docker** and deployed them to **AWS EC2** instances, improving deployment consistency and reducing setup time by 40% for new service instances.
- Contributed to the development and refinement of quantitative trading strategies by analyzing large datasets and building predictive models, achieving a **0.82 AUC** on a simulated trading environment.

SKILLS

Programming Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Business Intelligence & Analytics – Power BI, Tableau, DAX, Power Query, Statistical Analysis

Tools & Methodologies – Jira, Confluence, Agile, Splunk, Dynatrace

PROJECTS

Systematic Trading Strategy Backtesting Engine

- Developed a Python-based backtesting engine to evaluate systematic trading strategies using historical market data, simulating over 10,000 trading days.
- Implemented feature engineering pipelines in Python using **pandas** to extract relevant signals from diverse data sources, improving model performance by **15%**.
- Trained and evaluated a **scikit-learn** regression model to predict asset price movements, achieving a **Mean Absolute Error of 0.045** on out-of-sample data.

Real-Time Data Feed Processing Pipeline

- Constructed a scalable Python data processing pipeline on **AWS Lambda** to ingest and transform real-time market data streams, handling up to 2,000 events per second.
- Integrated the pipeline with **PostgreSQL** for storing processed data, enabling low-latency access for downstream analytical tools and trading simulations.
- Designed monitoring and alerting mechanisms using **CloudWatch** to ensure data integrity and pipeline operational status, reducing data latency by **30%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)